JSC370 Final Project: Country-Level Cancer Mortality

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GitHub repository: https://github.com/Terrachizoid/JSC370Midterm

1 Introduction

Cancer mortality varies substantially across countries and is shaped by a complex combination of behavioral, environmental, demographic, economic, and healthcare-related factors. Understanding whether broad country-level indicators are associated with these differences can help identify which factors are most informative at the population level, while also highlighting the limitations of using aggregated international health data for prediction.

This project examines cross-country variation in cancer mortality using publicly available health and development indicators. The main data source is the **World Health Organization (WHO) Global Health Observatory API**, which provides internationally comparable measures of cancer mortality and several major risk factors. To supplement these data, the **World Bank API** is used to obtain GDP per capita and life expectancy at birth. Together, these sources allow the construction of a country-level dataset combining mortality, lifestyle risk factors, environmental exposure, socioeconomic conditions, and healthcare resources.

The original goal was to build a recent country-year panel and study changes over time. However, after retrieving the indicators, it became clear that year coverage was highly uneven across variables and countries. Some indicators were reported regularly, while others appeared only sporadically within the target period. To obtain a dataset with broader country coverage and a common set of predictors, the analysis was therefore reformulated as a **cross-sectional study**, using the most recent available observation for each country between 2014 and 2023.

The main research question is: **To what extent can country-level lifestyle risk factors, socioeconomic conditions, environmental exposure, demographic structure, and healthcare resources explain variation in cancer mortality across countries?**

I hypothesized that smoking, obesity, alcohol consumption, and physical inactivity would be positively associated with cancer mortality. I also expected life expectancy to be related to mortality because it partly reflects population age structure, longevity, and overall health-system development. At the same time, I expected predictive performance to be limited, since cancer mortality is influenced by many overlapping processes that are not fully captured by a small set of aggregate country-level indicators.

2 Data and Preprocessing

2.1 Data sources and study design

Data were collected from two public APIs. The **WHO Global Health Observatory API** was used to retrieve the outcome variable, age-standardized cancer mortality, along with several candidate predictors: smoking prevalence, obesity prevalence, alcohol consumption, prevalence of insufficient physical activity, PM2.5 exposure, and medical doctors per 10,000 population. The **World Bank API** was used to obtain GDP per capita and life expectancy at birth. These variables were selected to represent behavioral risk factors, environmental exposure, socioeconomic development, demographic structure, and healthcare capacity.

Because the temporal coverage of the indicators was inconsistent across countries and variables, the analysis was designed as a **cross-sectional country-level study** rather than a panel analysis. For each variable, observations were restricted to the 2014–2023 period, and the most recent available value for each country was retained. The resulting country-level records were then merged into a single analytic dataset.

2.2 Data acquisition and preprocessing

The code below imports the required packages and defines helper functions used to download, filter, and standardize the indicators from the WHO and World Bank APIs.

Table 1 summarizes the indicators included in the final analytic dataset.

	variable	source	indicator_code	description
0	cancer_mortality	WHO	SA_0000001807	Age-standardized cancer mortality rat
1	smoking	WHO	M_Est_tob_curr	Estimated current tobacco use prevale
2	obesity	WHO	NCD_BMI_30A	Adult obesity prevalence (BMI >= 30
3	alcohol	WHO	SA_0000001822	Alcohol consumption indicator
4	inactivity	WHO	NCD_PAA	Insufficient physical activity prevalenc
5	gdp_per_capita	World Bank	NY.GDP.PCAP.CD	GDP per capita (current US$)
6	pollution_pm25	WHO	SDGPM25	PM2.5 exposure indicator
7	medical_doctors_per_10000	WHO	HWF_0001	Medical doctors per 10,000 population
8	life_expectancy	World Bank	SP.DYN.LE00.IN	Life expectancy at birth (years)

2.3 Building the country-level dataset

The selected indicators were downloaded and restricted to the 2014–2023 period. Because year coverage differs across indicators, the most recent available observation for each country in the selected window was retained.

This produces a cross-sectional dataset rather than a sparse country-year panel.

	variable	countries_available
0	cancer_mortality	180
1	smoking	167
2	alcohol	194
3	obesity	199
4	inactivity	195
5	gdp_per_capita	262
6	pollution_pm25	195
7	medical_doctors_per_10000	193
8	life_expectancy	262

Cancer mortality was used as the base table because it is the outcome of interest. Predictor tables were then merged by country.

	variable	missing_before_imputation
0	country	0
1	cancer_mortality	0
2	smoking	22
3	alcohol	0
4	obesity	0
5	inactivity	0
6	gdp_per_capita	5
7	life_expectancy	0
8	pollution_pm25	0
9	medical_doctors_per_10000	0

2.4 Missing data and analytic variables

Remaining numeric missing values were imputed using variable medians. This is a simple and transparent strategy that preserves country coverage for both exploratory analysis and predictive modeling while avoiding a substantial loss of observations from complete-case analysis.

	variable	missing_after_imputation
0	country	0
1	cancer_mortality	0
2	smoking	0
3	alcohol	0

	variable	missing_after_imputation
4	obesity	0
5	inactivity	0
6	gdp_per_capita	0
7	life_expectancy	0
8	pollution_pm25	0
9	medical_doctors_per_10000	0

The final analytic dataset contains one row per country. Table 2 reports summary statistics for the numeric variables.

	variable	mean	median	std	min	max
0	Cancer mortality	122.89	118.09	37.78	48.35	262.45
1	Smoking prevalence (%)	19.34	17.40	13.97	0.00	74.70
2	Alcohol consumption	6.01	4.15	5.41	0.00	24.60
3	Obesity prevalence (%)	22.27	22.20	12.62	1.06	62.43
4	Physical inactivity prevalence (%)	27.01	23.76	13.90	2.94	72.03
5	GDP per capita (current US$)	16776.50	6299.99	23307.51	250.63	133230.62
6	Life expectancy at birth (years)	73.11	73.60	6.98	54.46	84.16
7	PM2.5 exposure	22.86	17.18	14.85	5.47	67.22
8	Medical doctors (per 10,000)	20.26	16.76	17.30	0.38	85.05

3 Exploratory Data Analysis

Exploratory data analysis included histograms, scatterplots, correlation summaries, and boxplots to assess variable distributions, inspect bivariate relationships with cancer mortality, and identify skewness or outliers that might affect modeling. Interactive visualizations were also created as a companion website component to support geographic interpretation, country comparison, and exploration of selected predictors.

3.1 Distribution of variables

Figure 1 shows the distributions of the numeric variables in the cleaned dataset.

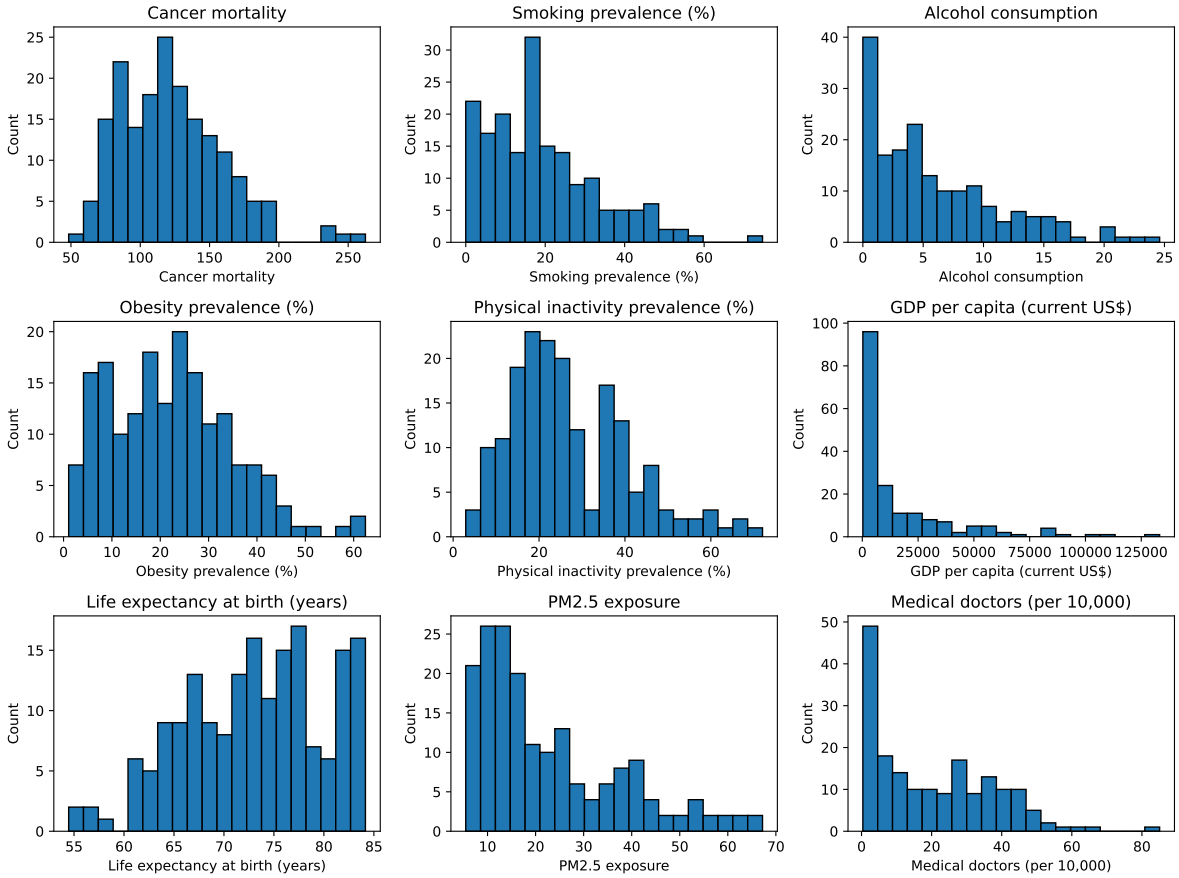


Figure 1: Figure 1. Distributions of cancer mortality and predictor variables across countries. GDP per capita is especially right-skewed, while several other variables also show noticeable cross-country heterogeneity.

Most variables show some degree of right-skewness. GDP per capita is the clearest example, reflecting the large gap between lower-income and higher-income countries, while life expectancy appears more concentrated with less extreme spread. This suggests that transformations such as a log scale may be useful for selected predictors in the predictive modeling stage.

3.2 Bivariate relationships with cancer mortality

Figure 2 shows scatterplots of cancer mortality against the main predictors.

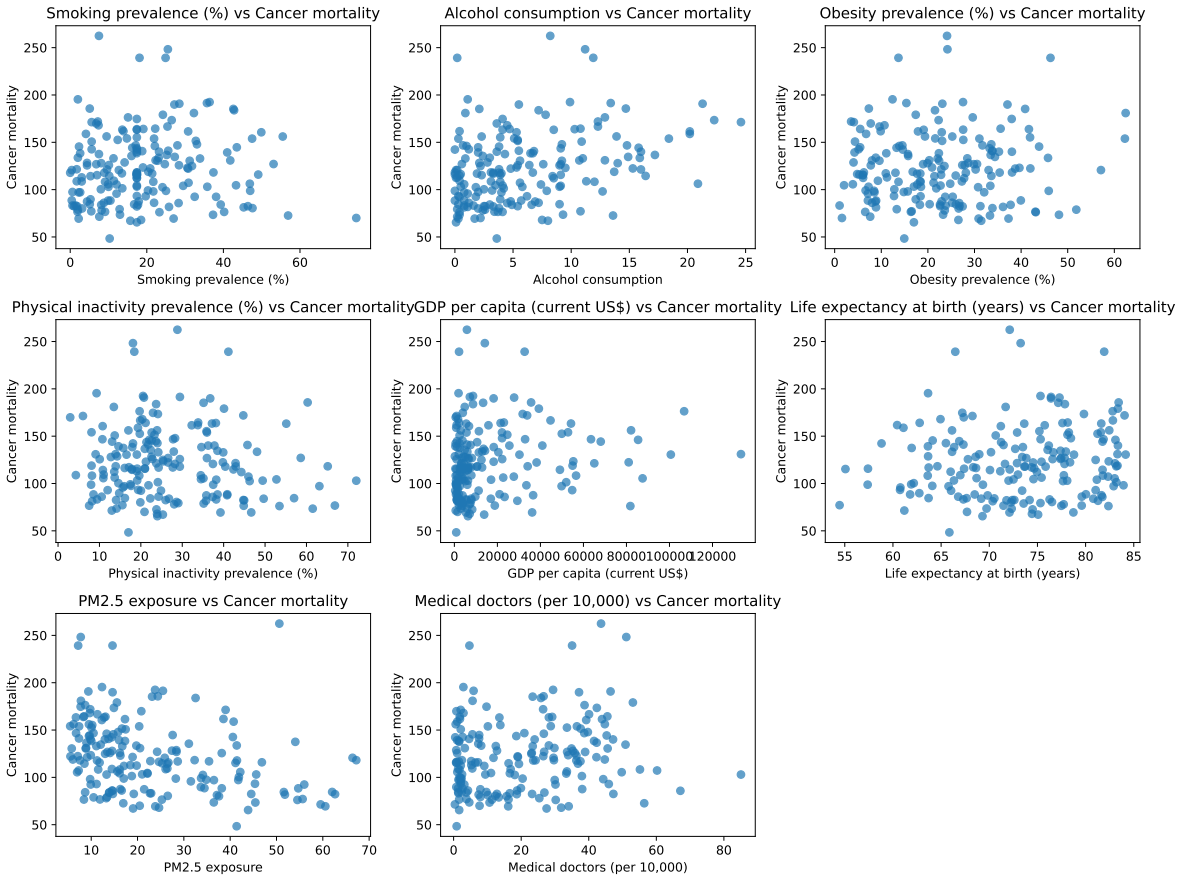


Figure 2: Figure 2. Scatterplots of cancer mortality against selected predictors. Most relationships appear weak to moderate and are accompanied by substantial variability, suggesting that no single variable explains country-level cancer mortality particularly well on its own.

The scatterplots suggest weak to moderate relationships overall. Smoking, alcohol consumption, and obesity show slight upward patterns, while life expectancy appears to capture broader demographic and development differences that may also matter for cancer mortality. Even so, each pattern is accompanied by considerable noise. This is not surprising in a country-level analysis, where cancer mortality is shaped by many overlapping demographic, behavioral, environmental, and healthcare factors.

3.3 Correlation structure

Figure 3 presents the correlation matrix for the numeric variables.

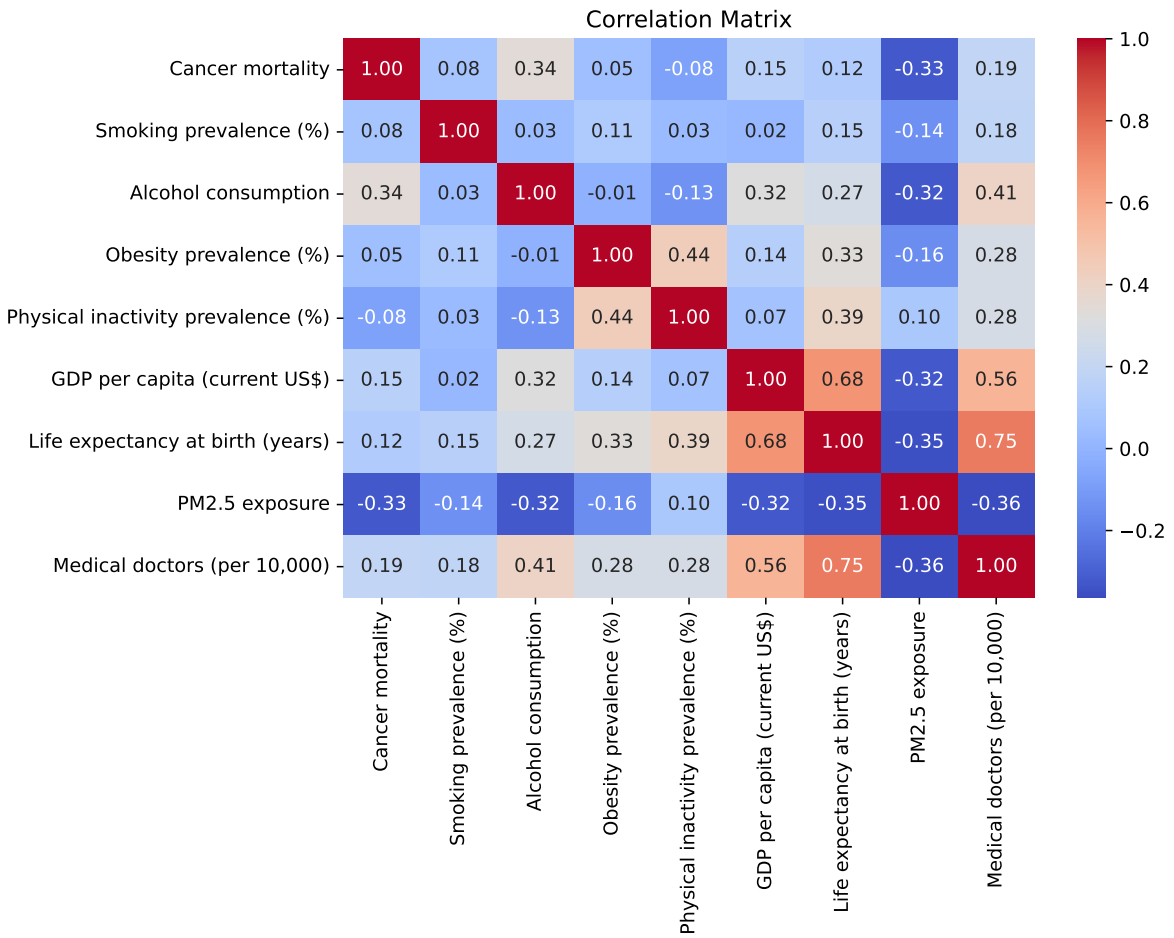


Figure 3: Figure 3. Correlation matrix of the numeric variables. Most correlations with cancer mortality are modest, while some stronger associations appear among the predictors themselves, such as GDP per capita and physician density.

The correlation matrix reinforces the same general pattern. Correlations between cancer mortality and individual predictors are mostly modest, although smoking, alcohol consumption, and obesity tend to have small positive associations. Life expectancy also appears as a broader population-health measure rather than a strong standalone driver. GDP per capita is positively related to physician density, which is consistent with the idea that wealthier countries tend to have greater healthcare capacity.

One noteworthy result is the negative correlation between PM2.5 exposure and cancer mortality. This should not be interpreted as evidence of a protective effect. A more plausible explanation is confounding from factors such as life expectancy, age structure, and differences in reporting quality across countries.

3.4 Outlier inspection

Figure 4 shows boxplots for the numeric variables.

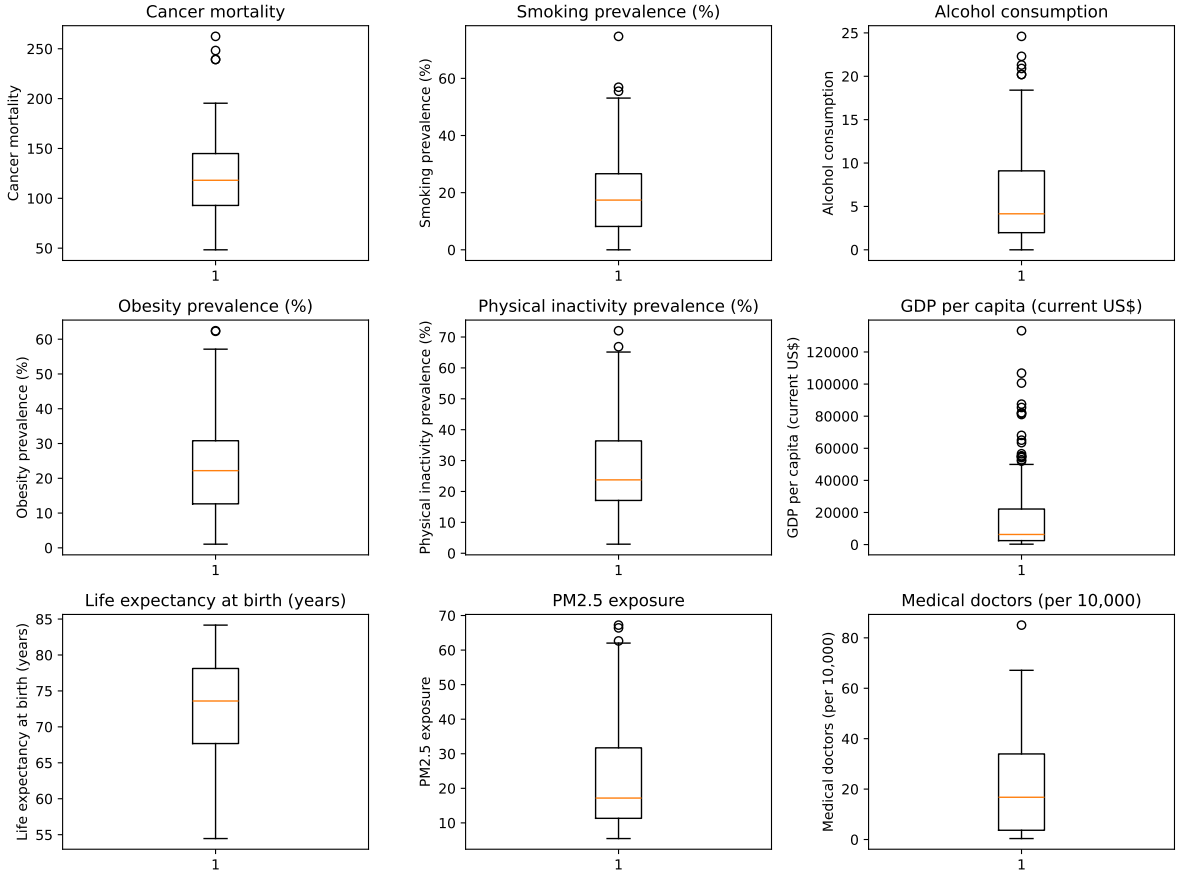


Figure 4: Figure 4. Boxplots for cancer mortality and predictor variables. Several variables, especially GDP per capita and physician density, contain high-end outliers, which may motivate transformations or robust modeling choices in the final project.

Several predictors contain noticeable high-end outliers, especially GDP per capita and physician density. This reflects the wide gap between countries rather than obvious data entry errors, but it suggests that careful treatment of skewed variables will matter in later modeling.

3.5 Interactive visualizations

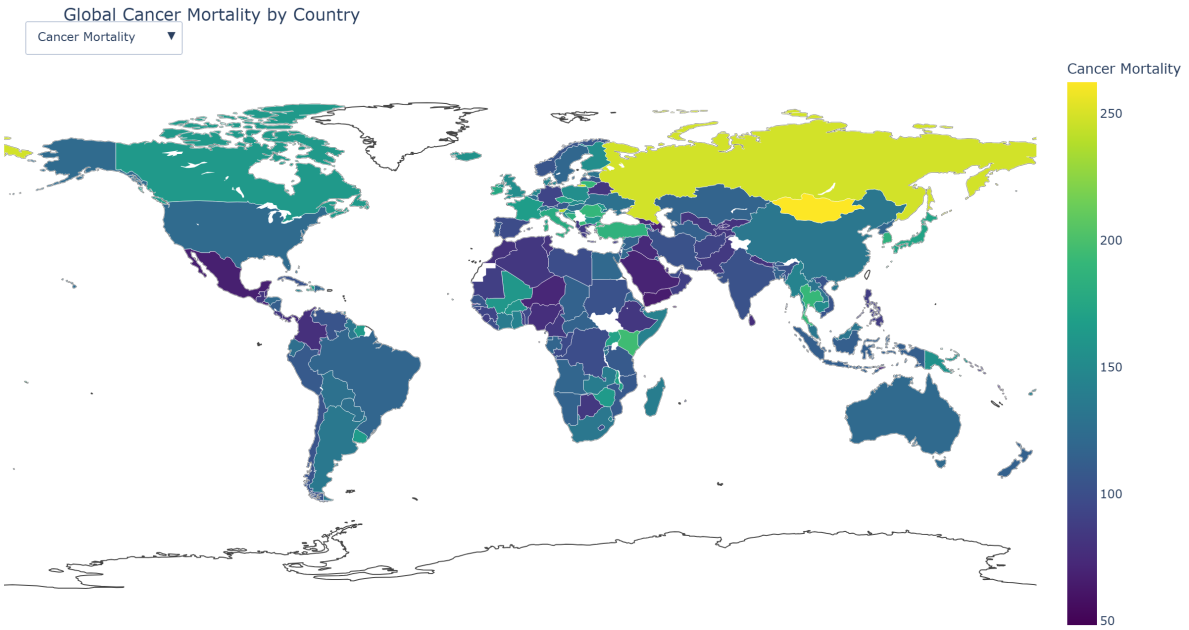
The following interactive figures extend the exploratory analysis using the cleaned country-level dataset developed for the project. They are presented in report style, with each visualization

paired with a short caption and interpretation. The goal is not to introduce new modeling results, but to give readers multiple interactive ways to examine the geographic distribution of the data, the relationship between selected predictors and cancer mortality, and the countries that rank highest or lowest on each measure.

The PDF version shows static versions of the interactive website figures so that the report remains readable in print. The full interactive versions are available on the website.

3.5.1 Interactive Figure 1. Global choropleth map of selected indicators

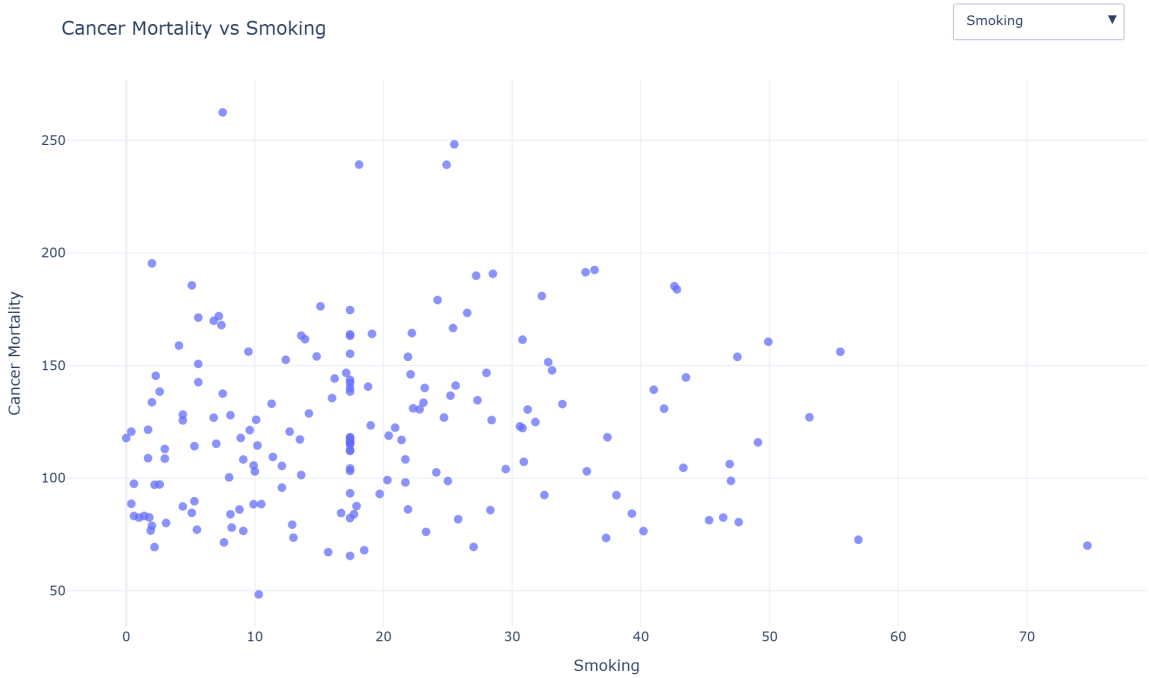
This figure provides a geographic view of the dataset. The dropdown menu allows the user to switch between cancer mortality and several predictors, including smoking prevalence, alcohol consumption, obesity prevalence, physical inactivity, GDP per capita, and life expectancy. Hover text displays the country name, ISO-3 code, and the selected value. This plot is useful for identifying broad regional structure and for seeing whether countries with high mortality also tend to score highly on the same predictors.



Caption. *Interactive choropleth map of cancer mortality and selected country-level predictors.* Use the dropdown menu to switch the displayed variable. Darker shading indicates larger values for the selected measure, and hovering over a country reveals its name, ISO-3 code, and numeric value.

3.5.2 Interactive Figure 2. Cancer mortality versus selected predictors

This scatter plot fixes cancer mortality on the y-axis and lets the reader choose a single predictor for the x-axis. That design is intentionally simpler than a two-dropdown version: it avoids synchronization problems and keeps the interpretation focused on the main project question, namely how each candidate predictor relates to cancer mortality. Each point is a country, and hovering reveals the country name, ISO-3 code, and both variable values.



Caption. *Interactive scatter plot of cancer mortality against selected predictors.* The dropdown menu changes the predictor shown on the x-axis while cancer mortality remains fixed on the y-axis. This view makes it easier to compare whether smoking, alcohol consumption, obesity, physical inactivity, GDP per capita, or life expectancy show stronger or weaker country-level associations with mortality.

3.5.3 Interactive Figure 3. Country rankings for selected indicators

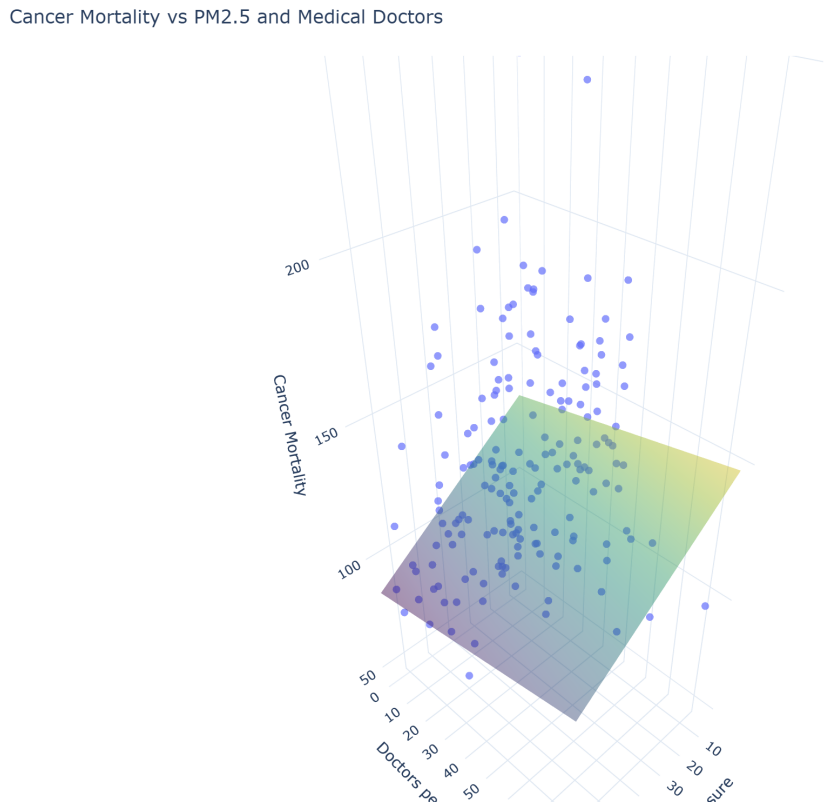
The third figure is an interactive ranking table. It allows the user to switch between variables and inspect the countries with either the highest or lowest observed values. This complements the map and scatter plot by giving an exact lookup of country names and values rather than only a visual impression.

	Rank	Country	ISO-3	Cancer Mortality
0	1	Mongolia	MNG	262.448127
1	2	Russian Federation	RUS	248.271735
2	3	Slovenia	SVN	239.270063
3	4	Kiribati	KIR	239.221815
4	5	Kenya	KEN	195.388000
5	6	North Macedonia	MKD	192.473933
6	7	Thailand	THA	191.464763
7	8	Lithuania	LTU	190.767123
8	9	Romania	ROU	189.912971
9	10	Korea, Republic of	KOR	185.646510

Caption. *Interactive ranking table for country-level indicators.* The dropdown menu switches between variables and ordering rules, allowing the reader to inspect the countries with the ten highest or ten lowest values for the selected measure.

3.5.4 Exploratory figure. 3D scatter plot with fitted plane

This final figure is not part of the three required HW5 visuals, but it is included as an exploratory extension. It plots cancer mortality against PM2.5 exposure and medical doctor density, then overlays a fitted linear plane. The purpose is not to claim a causal mechanism or to present a final predictive model, but to illustrate that some of the most surprising bivariate findings become easier to discuss when viewed jointly.



Caption. *Exploratory 3D scatter plot of cancer mortality, PM2.5 exposure, and medical doctor density with fitted linear plane.* The fitted plane indicates a positive association between doctor density and recorded cancer mortality, alongside a negative association between PM2.5 exposure and mortality. These patterns are counterintuitive and should not be interpreted causally; they more plausibly reflect confounding by development, healthcare access, life expectancy, and reporting quality.

4 Predictive Modeling

4.1 Modeling strategy

To extend the exploratory analysis into the predictive stage of the project, I compared three regression models: a baseline **ordinary least squares linear regression**, a **random forest regressor**, and an **XGBoost regressor**. The same outcome and predictor set were used across models so that performance differences would reflect model flexibility rather than differences in inputs.

The outcome variable was **cancer mortality**. The predictor set included smoking, obesity, alcohol consumption, physical inactivity, GDP per capita, PM2.5 exposure, medical doctor

density, and life expectancy. For the linear model, GDP per capita, PM2.5 exposure, and doctor density were log-transformed to reduce skewness. The tree-based models were fit on the original scale because they do not require linearity or normally distributed predictors.

Model performance was assessed with **5-fold cross-validation** using three metrics: **(R^2)**, **RMSE**, and **MAE**. To provide a useful benchmark, a null model that predicts the mean cancer mortality was also included in the final comparison table.

4.2 Baseline models

The linear regression model serves as an interpretable baseline, while the initial random forest model provides a constrained nonlinear benchmark with shallow trees and minimum node-size restrictions to reduce overfitting.

	Model	CV R^2	CV R^2 SD	CV RMSE	CV MAE	Train R^2
0	Null model	-0.0208	0.0276	37.6186	29.8565	NaN
1	Linear regression	0.1003	0.1069	35.3263	27.0053	NaN
2	Random forest (constrained)	0.1216	0.1139	34.9405	26.8325	0.4906

These baseline results provide a reference point for judging whether more flexible models meaningfully improve prediction or mainly increase variance.

4.3 Random forest tuning

Random forest hyperparameters were tuned in two stages. A small **randomized search** was first used to explore a broad but manageable parameter space. A more focused **grid search** was then conducted around the better-performing region suggested by the randomized search. This two-stage strategy is more efficient than a large exhaustive grid while still allowing the main bias–variance trade-offs to be explored.

	mean_test_score	std_test_score	mean_train_score	param_n_estimators	param_max_depth	p
8	0.1284	0.1165	0.4566	100	4	1
1	0.1225	0.0903	0.4350	100	3	1
9	0.1223	0.0936	0.5014	200	4	1
10	0.1221	0.1088	0.4889	100	4	5
14	0.1221	0.0885	0.4131	100	3	1
7	0.1214	0.1031	0.4869	200	4	5
17	0.1212	0.0872	0.4526	100	3	5
12	0.1198	0.1063	0.4479	200	4	5
11	0.1194	0.0956	0.3453	200	3	1

	mean_test_score	std_test_score	mean_train_score	param_n_estimators	param_max_depth	p
15	0.1192	0.1007	0.3828	300	4	5

	mean_test_score	std_test_score	mean_train_score	param_n_estimators	param_max_depth	p
0	0.1312	0.0962	0.4011	100	3	5
2	0.1294	0.0922	0.4072	300	3	5
1	0.1293	0.0912	0.4037	200	3	5
5	0.1275	0.0926	0.4013	300	3	1
39	0.1265	0.1089	0.4758	100	4	1
3	0.1262	0.0986	0.3953	100	3	1
4	0.1259	0.0919	0.3978	200	3	1
7	0.1239	0.0967	0.3718	200	3	5
10	0.1239	0.0967	0.3718	200	3	1
41	0.1239	0.1041	0.4749	300	4	1

	max_depth	max_features	min_samples_leaf	min_samples_split	n_estimators
0	3	0.5	4	5	100

The random forest tuning results indicate modest restrictions on tree depth, node size, and feature subsampling improve generalization.

Increasing the number of trees beyond 100 did not improve predictive performance. While random forests theoretically reduce variance as the number of trees increases, this effect diminishes once a sufficient number of trees has been reached. In this dataset, performance differences between 100, 200, and 300 trees were negligible relative to cross-validation variability, indicating that model performance had already plateaued.

Instead, hyperparameters controlling tree complexity and feature subsampling (e.g., maximum depth, minimum samples per leaf, and max_features) had a larger impact on generalization performance, likely by reducing correlation between trees and preventing overfitting.

4.4 XGBoost modeling and tuning

XGBoost was included as a second tree-based model with a different inductive bias. Unlike random forests, which average many independently grown trees, XGBoost builds trees sequentially so that each new tree attempts to correct the errors of the previous ensemble. This can improve performance when the underlying structure is complex, but it also makes the method more prone to overfitting on small or noisy datasets.

	Model	CV R^2	CV R^2 SD	CV RMSE	CV MAE	Train R^2
0	XGBoost (default)	0.0319	0.1362	36.5887	28.6872	0.9219

We are seeing the default XGBoost was overfitting with train accuracy 0.92.

A focused grid search was used for XGBoost rather than a very large search space. Given the small and noisy dataset, the search emphasized shallower trees, moderate numbers of boosting rounds, subsampling, and stronger regularization.

	mean_test_score	std_test_score	mean_train_score	param_n_estimators	param_max_depth	p
14	0.1164	0.1275	0.6919	100	3	0
51	0.1130	0.1170	0.5305	100	2	0
50	0.1112	0.1039	0.5280	100	2	0
2	0.1091	0.1103	0.5131	100	2	0
3	0.1044	0.1086	0.5179	100	2	0
49	0.1005	0.1028	0.5907	100	2	0
55	0.1003	0.1180	0.6225	150	2	0
61	0.0979	0.1234	0.7890	100	3	0
62	0.0955	0.1006	0.7079	100	3	0
54	0.0940	0.1131	0.6199	150	2	0

	colsample_bytree	learning_rate	max_depth	n_estimators	reg_lambda	subsample
0	0.7	0.05	3	100	5	0.7

4.5 Final model comparison

The final comparison table summarizes cross-validated performance across the null baseline, linear regression, constrained and tuned random forest models, and default and tuned XGBoost models.

	Model	Train R^2	CV R^2	CV R^2 SD	CV RMSE	CV MAE
0	Null model	NaN	-0.0208	0.0276	37.6186	29.8565
1	Linear regression	NaN	0.1003	0.1069	35.3263	27.0053
2	Random forest (constrained)	0.4906	0.1216	0.1139	34.9405	26.8325
3	Random forest (tuned)	0.4011	0.1312	0.0962	34.7598	26.9763
4	XGBoost (default)	0.9219	0.0319	0.1362	36.5887	28.6872
5	XGBoost (tuned)	0.6919	0.1164	0.1275	34.9930	27.5004

4.6 Model results and interpretation

The predictive performance of all models is summarized in Table X. Overall, none of the models achieved strong predictive accuracy, with cross-validated **(R^2)** values remaining close to zero across all approaches.

The linear regression model provided a simple baseline and achieved slightly positive but very limited explanatory power. More flexible models, including random forest and XGBoost, did not substantially improve performance. In fact, the default XGBoost model performed worse than the null model, indicating severe overfitting and poor generalization.

A consistent pattern across models is that increasing model complexity did not lead to meaningful improvements in predictive performance. While tree-based methods are capable of capturing nonlinear relationships, their added flexibility resulted primarily in increased variance rather than improved fit to the underlying signal. This is reflected in the large gap between training and cross-validation performance for more complex configurations.

After hyperparameter tuning, both random forest and XGBoost models showed improved generalization. The tuned models used shallower trees, larger minimum node sizes, and subsampling strategies, which effectively reduced overfitting. This is evident from the reduction in training performance accompanied by modest improvements in cross-validated metrics.

Among all models, the tuned random forest achieved the best overall performance, although the improvement over simpler models was small. This suggests that while some nonlinear structure may exist, it is weak and difficult to exploit reliably given the available data.

5 Conclusions and limitations

This analysis constructed a cross-country dataset linking cancer mortality to lifestyle, environmental, socioeconomic, demographic, and healthcare indicators using data from the WHO and World Bank APIs. While exploratory analysis revealed substantial variation across countries, the predictive modeling results indicate that these variables provide only limited explanatory power for cancer mortality.

Across all models, predictive performance remained low, with cross-validated **(R^2)** values close to zero. Even the best-performing model, a tuned random forest, showed only modest improvement over the null model. This suggests that the relationship between the selected predictors and cancer mortality is weak at the country level.

The modeling results also highlight an important bias–variance trade-off. More flexible models such as random forests and XGBoost introduced higher variance without substantially reducing bias, resulting in poorer generalization performance. Hyperparameter tuning improved this by enforcing simpler model structures, which reduced overfitting but did not meaningfully increase predictive accuracy.

These findings indicate that the main limitation is not the choice of model, but the data and problem formulation itself. Cancer mortality is influenced by complex and long-term processes, including historical exposure, age structure, healthcare systems, screening practices, and reporting differences. Aggregating these effects into a small set of country-level indicators likely obscures important variation and introduces substantial irreducible noise.

Several data limitations further constrain the analysis. The WHO API contains substantial missing data across countries and years, which required restricting the dataset to the most recent available observations. This reduced the effective sample size and removed temporal information. In addition, some key variables were unavailable or inconsistently reported, leading to the use of proxies such as physicians per population for healthcare capacity. Median imputation was used to handle missing values, which simplifies the data structure but may further reduce variability and signal.

Overall, the results suggest that predicting cancer mortality using a small set of aggregated country-level indicators is inherently challenging. The low predictive performance reflects both weak observable signal and high irreducible noise. As a result, this analysis is better interpreted as a descriptive comparison of broad patterns rather than a precise predictive or causal model.